

Quantum mechanics $\rightarrow$ Microscopic

Newton $\rightarrow$ Macroscopic

De Broglie wavelength

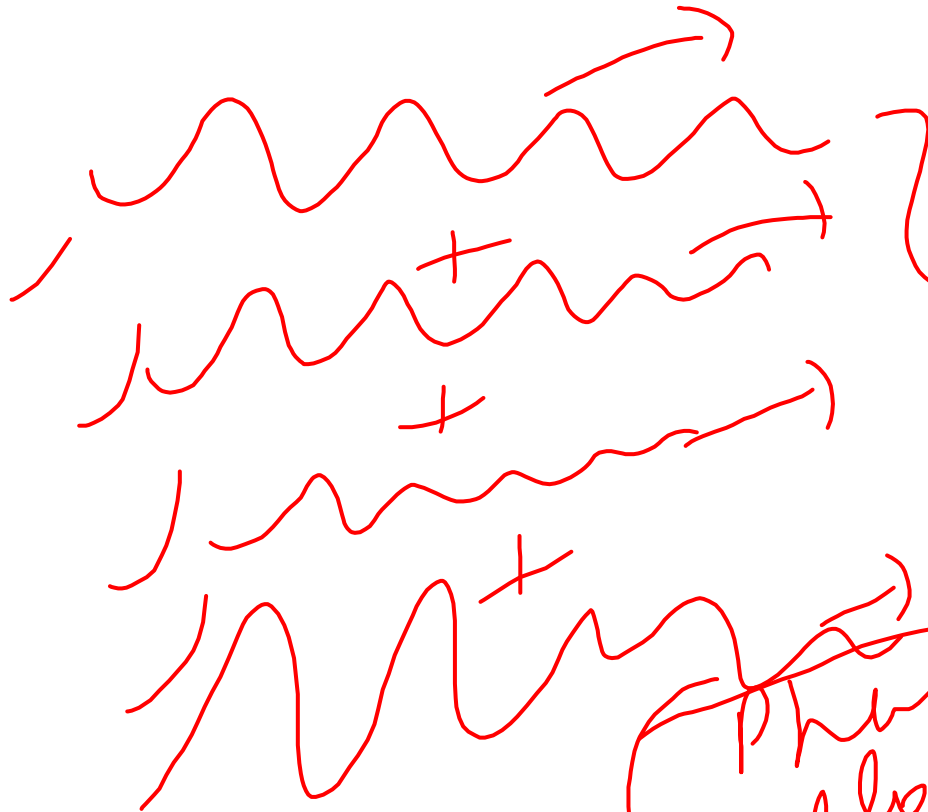
$$\lambda = \frac{h}{p}$$

Phase Velocity

$$V_p = \frac{c^2}{v}$$

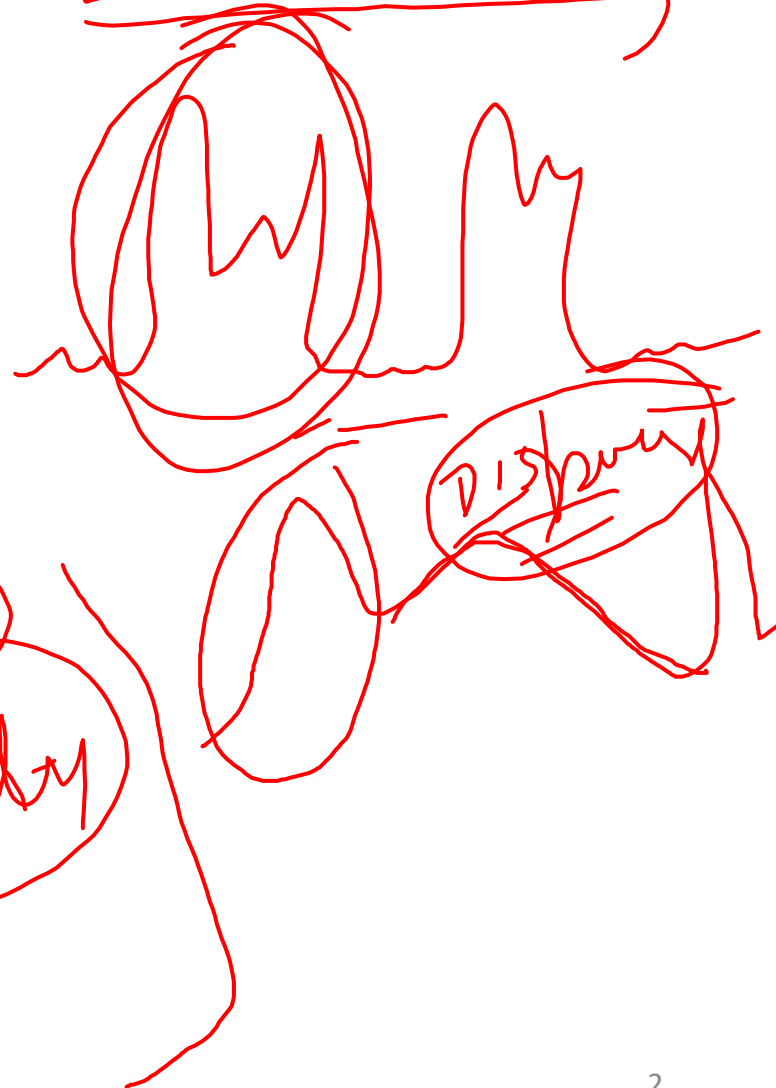
$$V_p = \frac{c^2}{v}$$

Group Velocity



Phase velocity

Group Velocity



Lecture #24

Content to be covered

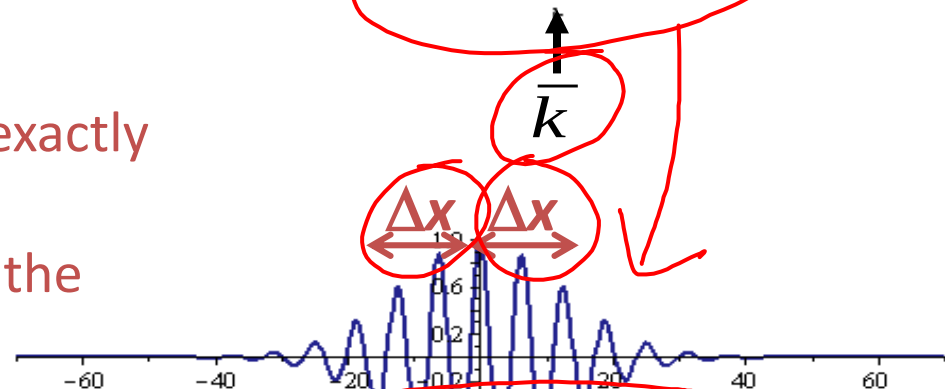
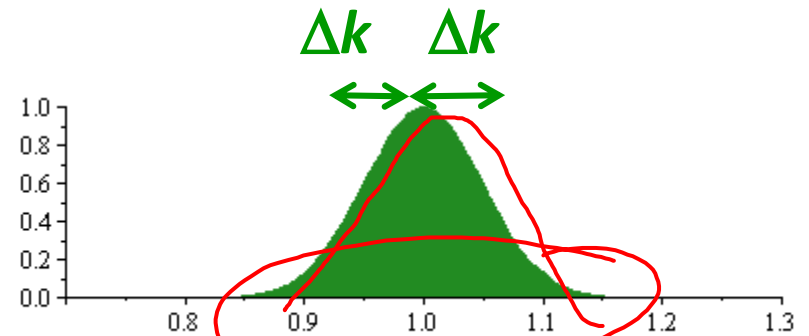
- ❖ Uncertainty Principle
- ❖ Applications of Uncertainty Principle
 - Heisenberg Microscope

Resources to be consulted

- Concepts of Modern Physics by Arthur Beiser

The Classical Uncertainty Principle

- The wave number of a wave packet is not exactly one value
 - It can be approximated by giving the central value
 - And the uncertainty, the “standard deviation” from that value
- The position of a wave packet is not exactly one value
 - It can be approximated by giving the central value
 - And the uncertainty, the “standard deviation” from that value



These quantities are related:

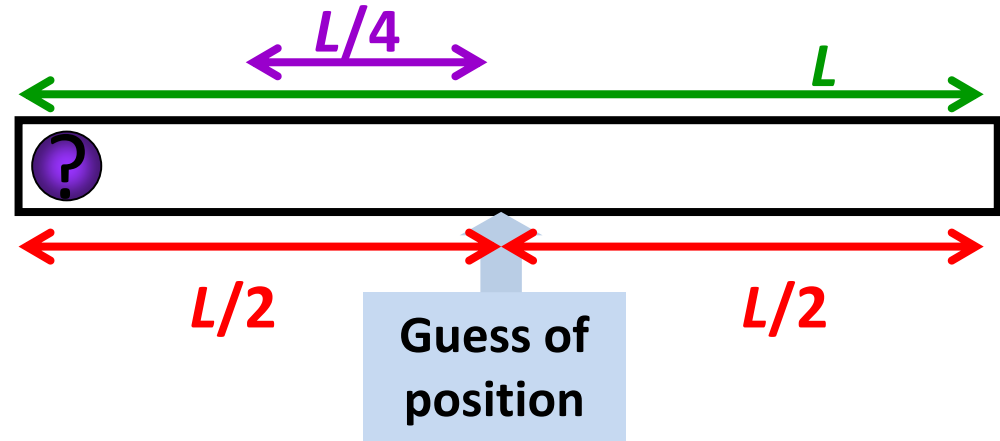
- Typically, $\Delta x \Delta k \sim 1$

$$\Delta x \Delta k \geq \frac{1}{2}$$

Estimating Uncertainty

A particle/wave is trapped in a box of size L

- What is the uncertainty in its position Δx ?

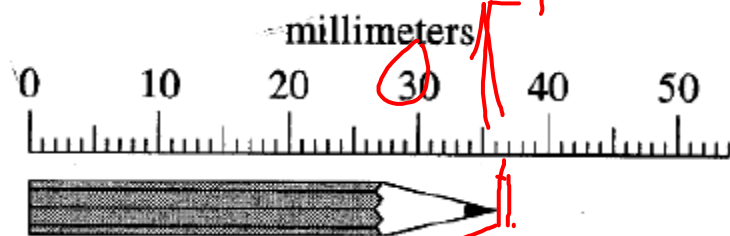


- Best guess: The particle is in the center, $x = L/2$
- But there is an error Δx on this amount
 - It is no greater than $L/2$
- It is certainly bigger than 0

Know the Error of Thy Ways: Measurement Error $\rightarrow \Delta$

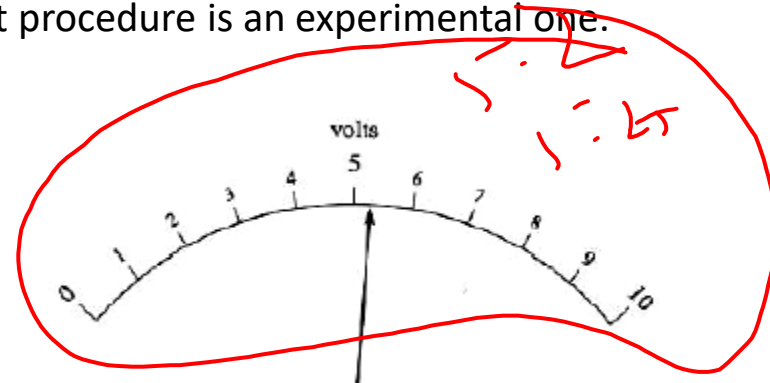
Measurements are made by observing something : length, time, momentum, energy

- All measurements have some (limited) precision...no matter the instrument used
- Examples:
 - How long is a desk ? $L = (5 \pm 0.1) \text{ m} = L \pm \Delta L$ (depends on ruler used)
 - How long was this lecture ? $T = (50 \pm 1) \text{ minutes} = T \pm \Delta T$ (depends on the accuracy of your watch)
 - How much your weight? $M = (1000 \pm 900) \text{ kg} = m \pm \Delta m$
- Is this a correct measure weight ?
 - Correct (because of large error reported) but imprecise
 - Correct weight is covered by the (large) error in observation
- Measurement errors are unavoidable since the measurement procedure is an experimental one.
- True value of an measurable quantity is an abstract concept.



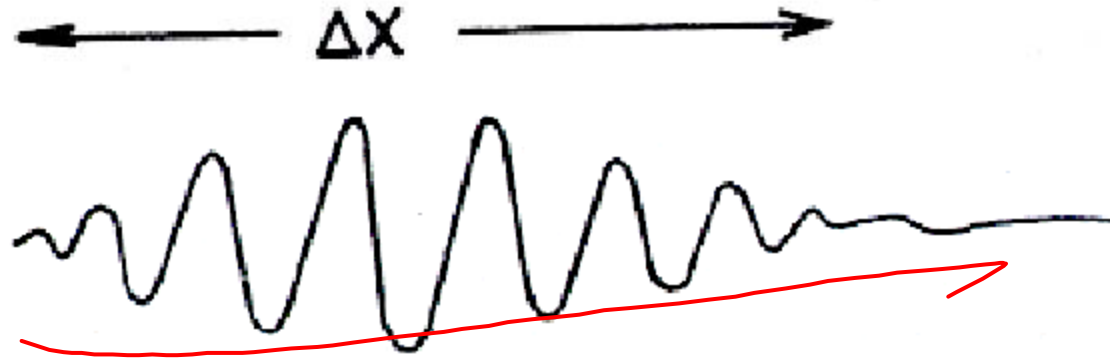
Best estimate of length: 36 mm
Probable range: 35.5 to 36.5 mm

Length Measurement



Best estimate of voltage: 5.3 V
Probable range: 5.1 to 5.4 V

Voltage (or time) Measurement



What is the Wave Length of this wave packet?

$$\underline{\lambda - \Delta\lambda} < \lambda < \underline{\lambda + \Delta\lambda}$$

De Broglie wavelength $\lambda = h/p$

→ Momentum Uncertainty: $\underline{p - \Delta p} < \underline{p} < \underline{p + \Delta p}$

Similarly for frequency ω or f

$$\omega - \Delta\omega < \omega < \omega + \Delta\omega$$

Planck's condition $E = hf = h\omega/2$

$$\rightarrow E - \Delta E < E < E + \Delta E$$

Heisenberg Uncertainty Principle

It is not possible to determine the **Position** and Linear **Momentum** of a particle with unlimited precision simultaneously.

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

If the measurement of the position of a particle is made with a precision Δx and a SIMULTANEOUS measurement of its momentum Δp in the X direction , then the product of the two uncertainties (measurement errors) can never be smaller than $\cong h/4\pi$ irrespective of how precise the measurement tools.

It is not possible to determine the **Energy** and **Time** of a particle with unlimited precision simultaneously.

$$\Delta E \Delta t \geq \frac{h}{4\pi}$$

It is not possible to determine the **Angular momentum** and **Angular position** of a particle with unlimited precision simultaneously.

$$\Delta J \Delta \theta \geq \frac{h}{4\pi}$$

Applications of Uncertainty Principle

Non existence of electron in the nucleus

Typical size (radius) of nucleus $\approx 10^{-14}$ m

$$\Delta p \Delta x \geq \frac{h}{2} \Rightarrow \Delta p \geq \frac{h}{2\Delta x}$$

For an electron to be confined inside Nucleus; $\Delta x \approx 10^{-14}$ m; So, $\Delta p \geq \frac{h}{2\Delta x}$; $\Delta p \approx \frac{1.054 \times 10^{-34}}{2 \times 10^{-14}}$

$$\Delta p \approx 1.054 \times 10^{-20} \text{ kg-ms}^{-1}$$

If this is the uncertainty in the electron's momentum, then its energy would be more than

$$E = \sqrt{c^2 p^2 + m_0^2 c^4}$$

Given: $p \approx 1.054 \times 10^{-20} \times 3 \times 10^8 = 3.3 \times 10^{-12} \text{ J} = 20.625 \text{ MeV}$ & $m_0 c^2 = 0.51 \text{ MeV}$

Thus: $E \geq 20 \text{ MeV}$.

Thus, it must have energy more than 20 MeV, if it is to be inside the nucleus.

Experimentally, it is observed that if an electron, which is associated with unstable atom never has energy more than a fraction of this energy. It concludes that electron cannot be present within the nucleus.

Applications of Uncertainty Principle

Non existence of electron in the nucleus

Assume that the electron is present in the nucleus of radius 10^{-14} m. for the existence of electron in the nucleus, the uncertainty Δx in its position must not be greater than 10^{-14} m. According to the uncertainty principle.

$$\begin{aligned}\Delta p &\geq h/2\pi\Delta x \\ &\geq 1.054 \times 10^{-34} \text{ Js} / 10^{-14} \text{ m} \\ &\geq 1.1 \times 10^{-20} \text{ kg-ms}^{-1}\end{aligned}$$

If this is the uncertainty in momentum of the electron then the momentum of the electron must be at least of the order of its magnitude, that is, $p \cong 1.1 \times 10^{-20} \text{ kg-ms}^{-1}$.

Then, velocity of electron $= p/m \approx 1.2 \times 10^{10} \text{ ms}^{-1}$ (impossible)

Now, if e^- is out side the nucleus then (assume) $\Delta x = 10^{-10} \text{ m}$

$$\Delta p \geq h/2\pi\Delta x \approx 1.1 \times 10^{-24} \text{ kg-ms}^{-1}$$

Then velocity for electron $= p/m \approx 1.2 \times 10^6 \text{ ms}^{-1}$ (possible)

Heisenberg Microscope

Suppose we have a particle, whose momentum is, initially, precisely known. For convenience assume initial $p = 0$.

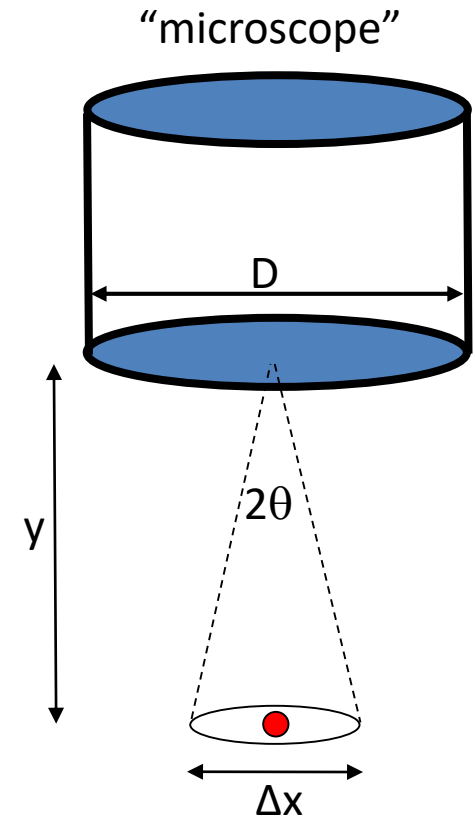
From wave optics (Rayleigh Criterion) $\sin \theta \approx \frac{\lambda}{D}$

From our diagram: $\sin \theta \approx \frac{\Delta x / 2}{y} = \frac{\Delta x}{2y}$

$$\Delta x \approx \frac{2y\lambda}{D}$$

Since this is a “thought experiment” we are free from any practical constraints, and we can locate the particle as precisely as we like by using radiation of shorter and shorter wavelengths.

But what are the consequences of this?



Heisenberg Microscope

In order to see the particle, a photon must scatter off it and enter the microscope.

Thus process MUST involve some transfer of momentum to the particle.....

BUT there is an intrinsic uncertainty in the X-component of the momentum of the scattered photon, since we only know that the photon enters the microscope somewhere within a cone of half angle α .

By conservation of momentum, there must be the same uncertainty in the momentum of the observed particle.....

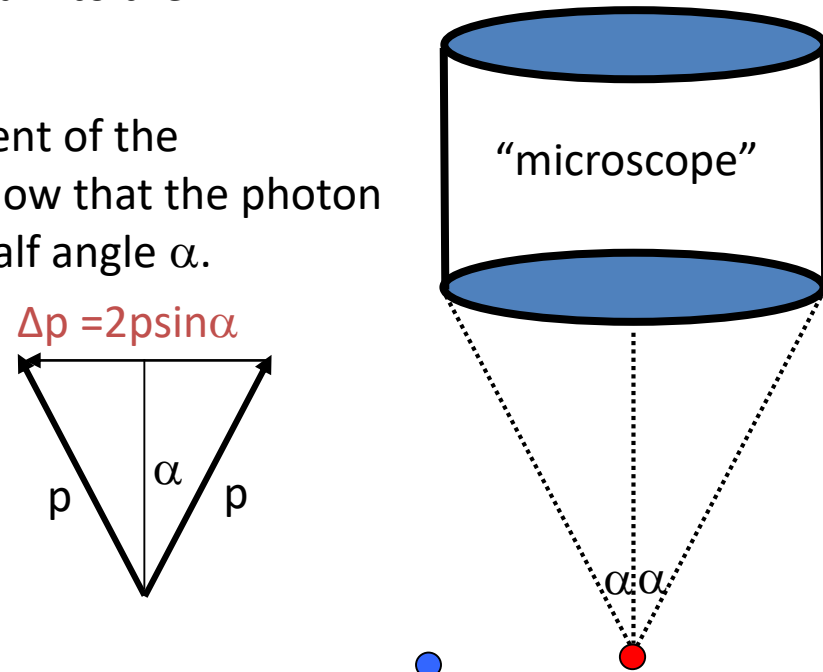
Uncertainty in position of particle: $\Delta x \approx \frac{2y\lambda}{D}$

Can reduce as much as we like by making λ small.....

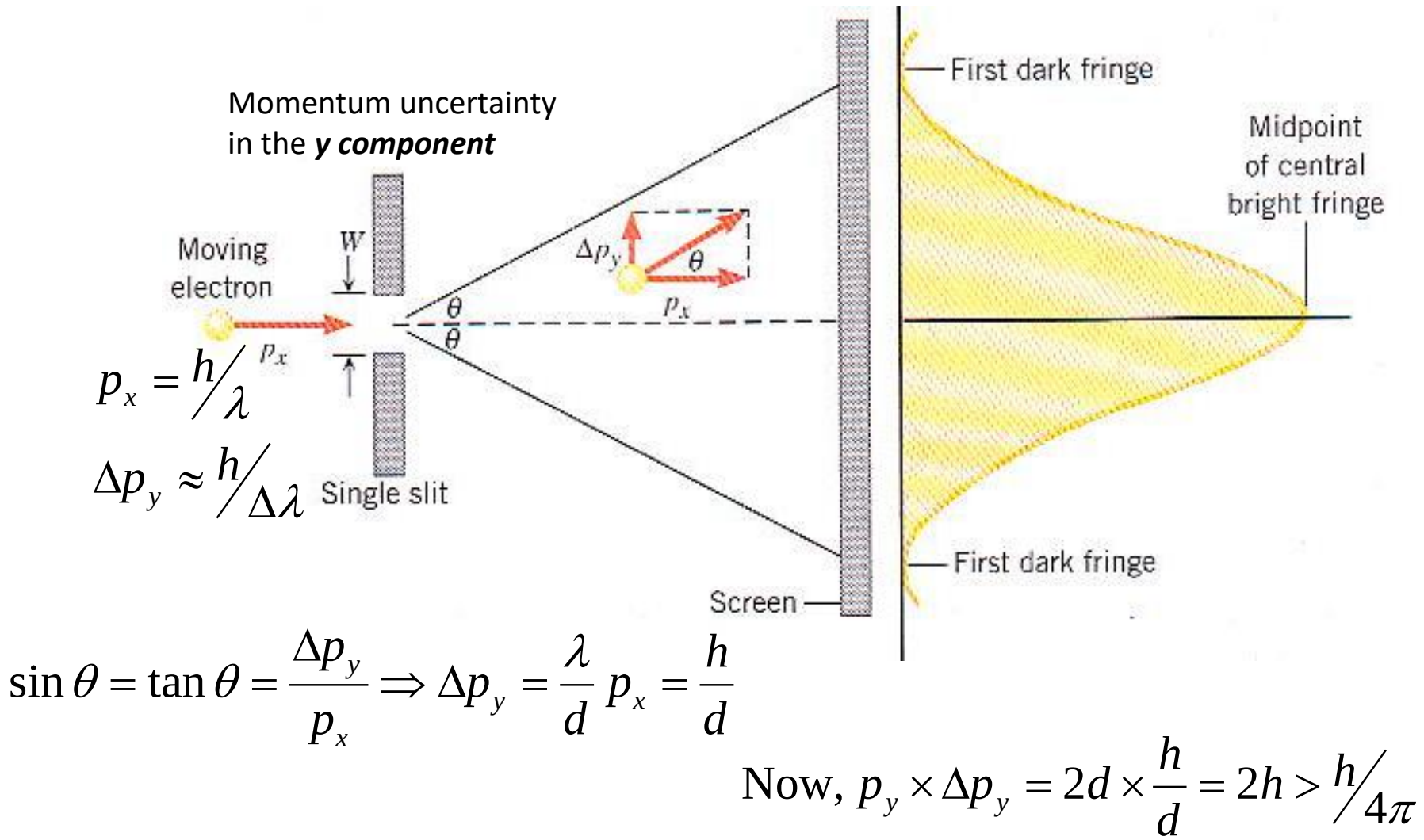
Uncertainty in momentum of particle: $\Delta p \approx 2p_{\text{photon}} \sin \alpha \approx \frac{2h}{\lambda} \frac{D}{2y}$

So, **if we attempt to reduce uncertainty in position by decreasing λ , we INCREASE the uncertainty in the momentum of the particle!!!!!!** Product of the uncertainties in position and momentum given by:

$$\Delta x \Delta p \approx \frac{2y\lambda}{D} \frac{Dh}{y\lambda} = 2h$$

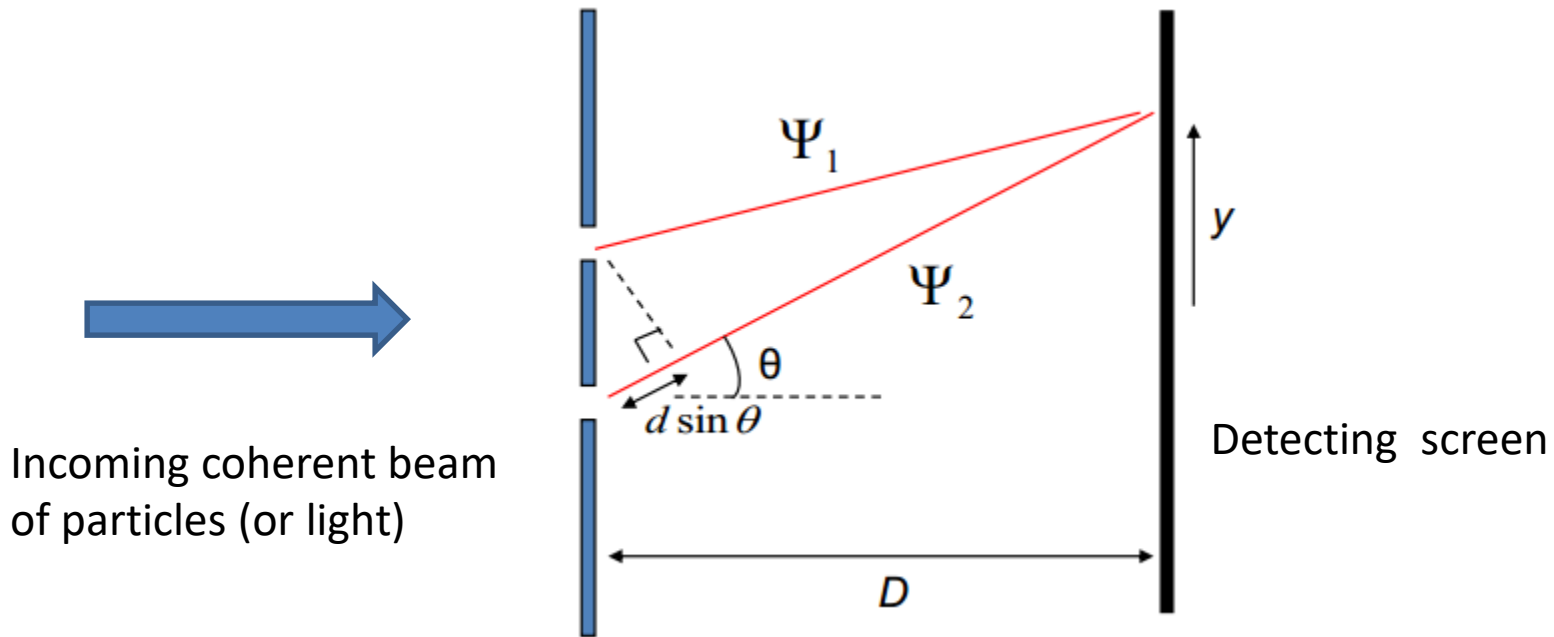


An illustration of Uncertainty Principle: Single slit diffraction



For single slit diffraction, $\sin \theta = \lambda / d$, postulates that λ is the de Broglie wavelength.

Double-slit Experiment Revisited



Schrödinger equation is linear: solution with both slits open is $\psi = \psi_1 + \psi_2$

Observation is nonlinear

$$|\psi|^2 = |\psi_1|^2 + |\psi_2|^2 + \psi_1^* \psi_2 + \psi_1 \psi_2^*$$

Usual “particle”
part

Interference term
gives fringes